KUO, CHEN CHIH 郭晨志

Chemical & Materials R&D Engineer • Battery Specialist

Seeking R&D Engineer, batteries & materials

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Xinzhuang, New Taipei · Open to relocation / shifts

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry, ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Communications** submission.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project executor**

Screening methodology
Designed full-stage test protocols, experimental workflows and statistical validation models.

Advanced analysis
EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Database & SOP
Established the test database and standardized high-throughput screening procedures (SOP).

Training & tech transfer
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Communications

Journal submission — battery thermal stability study

Status Under review **Co-author** Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Experience

2025 – Present — **Sustainable Electrochemical Energy Development (SEED) Center, NTUST**

Lab Equipment Procurement & Asset Maintenance Manager

- **Procurement (>NTD 2M):** Led procurement of glovebox O_2 sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables.
- **Lab safety:** monitor glovebox O_2 / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
- **Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li–$SOCl_2$ cells.
- Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures.
- **TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.
- **Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.

Summer 2022 — **New Taipei City Dept. of Urban Development**

Summer Intern, Housing Development Section

- Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes.

2021 – 2024 — **Tunghai University Student Housing / GEPT**

Resident Assistant (2 yrs) / Exam Staff (part-time)

- Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors.

Conferences & Training

Presenter **2026 TwIChE Annual Meeting**
2026/12/05–06 • Feng Chia University
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li ‖ SPAN Batteries: Solvation Structure, Transport Properties and Separator Wettability*

Vendor seminar **TA Instruments ARES-G3 launch & applications forum**
2026/07/17 • National Taiwan University

Vendor lecture **TA Instruments TGA lecture**
2026/06/03 • National Taiwan University

Skills & Instruments

Electrochemistry & batteries
Impedance spectroscopy (EIS) · Battery aging analysis · Reliability analysis · Low-temp electrolytes

Characterization
Thermal analysis (TGA / DSC) · Synchrotron (TLS 20A1 / TPS 27A2 / TPS 45A2) · Rheology (ARES-G3) · Viscometer (DMA 5002/Lovis 2001)

Process & operations
Lithography (nanoimprint / self-assembly) · Equipment procurement & maintenance

Software
Aspen Plus V14 · Aspen Dynamics · Microsoft Office · OriginPro · TA Universal Analysis · TA TRIOS

Education

2025 – 2027 (expected)
National Taiwan University of Science and Technology
M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

2020 – 2024
Tunghai University
B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B) — 2026 / 06 · certified · No. GB040231
- Indoor Air Quality Maintenance Management Specialist — 2026 / 08 · certified · No. IA020236
- Waste Disposal Technician (Class A) — 2026 / 10 · certified · No. HA160629
- Occupational Safety and Health Management Personnel (Level B) — 2026 / 11 · mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A) — 2027 / 03 · exam scheduled

Instruments

Maintenance
TGA Q500 · DSC 2500

Operation
Anton Paar DMA 5002/Lovis 2001 · ARES-G3 · Glovebox

Procurement
Glovebox O_2 sensor · DSC sample press · Hydraulic electrode punch · Vacuum pump · Eppendorf Pipette · Low-temperature bath